

NanoPitch *Coach*

A browser-based singing coach
that turns one recorded take
into objective, four-axis
feedback.

Built on the NanoPitch pitch / VAD model

THE PROBLEM

Beginner singers can't tell what changed between takes.

Was the pitch stable? Was the timing consistent?

Did the dynamics move at all? We wanted immediate, objective feedback after a short recording — not a replacement for a vocal coach.

WHAT IT DOES

Record one take. Read four axes of feedback.



Pitch

median note · f0 · voiced % · stability

Median note

A4

Voiced

84%



Tempo

onsets · BPM

Estimated BPM

Onsets



Dynamics

loudness contour · range used

Average dBFS

-31

Range used

13.8 dB



Technique

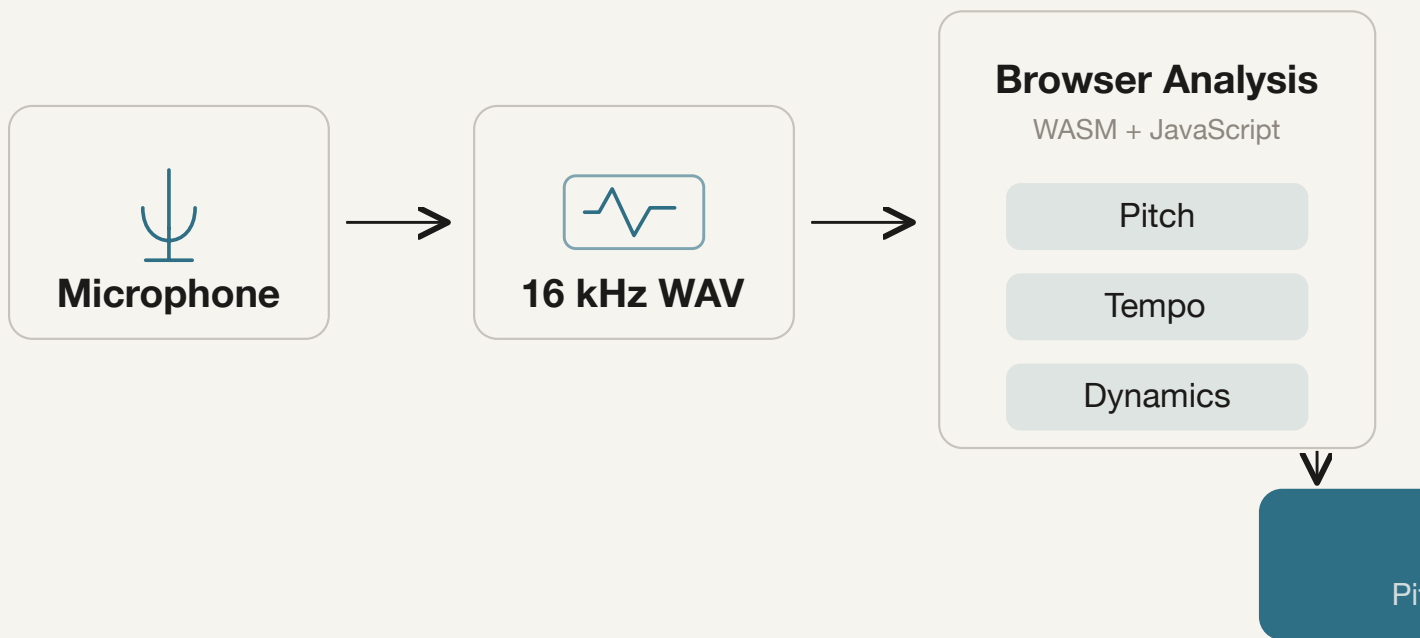
family · confidence

Detected family

Confidence

HOW IT WORKS

One recording, one pipeline.



RESULTS

Noise training made voice detection far more reliable.

Voice detection accuracy

600-clip held-out NanoPitch test set

Baseline



Random-SNR augmented



Improvement

+8.1 pts

–5 dB median pitch error

73.2 → 47.1

cents

LIMITATIONS & NEXT STEPS

The technique axis is honest about being demo-grade.

Submitted model

GT Singer English checkpoint

Best epoch	76
Clip accuracy	65.8%
Technique macro F1	0.497

Demo-grade fourth axis

needs

Next t

Better ta

- Mor
- Hum
- Mor

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